

Combination Calculation: 26 Choose 3



The weighted average formula is:

$$\text{Weighted Average} = \Sigma(\text{value} \times \text{weight}) / \Sigma(\text{weight})$$

Or written out: **Weighted Average** = $(v_1 \times w_1 + v_2 \times w_2 + v_3 \times w_3 + \dots) / (w_1 + w_2 + w_3 + \dots)$

Where:

- v = each value
- w = the corresponding weight for each value

The key difference from a regular average is that some values contribute more to the final result based on their assigned weights.

H

when do we use weighted average

Weighted averages are used when different data points have different levels of importance or influence. Here are common situations:

Academic Settings:

- Calculating final grades when exams, homework, and projects have different point values
- GPA calculations where courses have different credit hours

Financial Analysis:

- Portfolio returns where investments have different dollar amounts
- Cost of capital calculations weighted by debt and equity proportions
- Stock market indices (like S&P 500) weighted by market capitalization

Business Decisions:

- Employee performance reviews with different criteria having different importance
- Supplier evaluations where cost, quality, and delivery time matter differently
- Customer satisfaction scores when some customers represent more revenue

Statistical Analysis:

- Survey data where responses need to be weighted to match population demographics
- Quality control when sample sizes vary across different batches
- Research studies where some data points are more reliable than others

Everyday Examples:

- Restaurant ratings where recent reviews matter more than old ones
- Product reviews where verified purchases carry more weight
- Sports rankings considering strength of schedule

The key principle is: use weighted averages when a simple average would be misleading because not all data points should contribute equally to the final result.

H

why is the formula diff from normal avg. Why divide by weights

Great question! The division by weights is necessary to maintain the mathematical properties of an average.

Normal Average Logic:

- You add all values and divide by the COUNT of items
- Each item contributes equally (weight = 1 for each)
- Formula: $(v_1 + v_2 + v_3) / 3$

Weighted Average Logic:

- You add all weighted values but now divide by the TOTAL WEIGHT
- Items contribute differently based on their weights
- Formula: $(v_1 \times w_1 + v_2 \times w_2 + v_3 \times w_3) / (w_1 + w_2 + w_3)$

Why divide by total weights?

Think of it this way: if you have 3 test scores but one test is worth 2x the others:

- Test 1: 80 (weight = 1)
- Test 2: 90 (weight = 1)
- Test 3: 70 (weight = 2)

This is really like having FOUR test scores: 80, 90, 70, 70

So the average should be: $(80 + 90 + 70 + 70) / 4 = 77.5$

Using weighted formula: $(80 \times 1 + 90 \times 1 + 70 \times 2) / (1 + 1 + 2) = 310 / 4 = 77.5$

The key insight: Dividing by total weights accounts for the fact that some values are "counted multiple times" due to their higher importance. Without this adjustment, you'd get a distorted result that doesn't properly reflect the relative importance you assigned to each value.